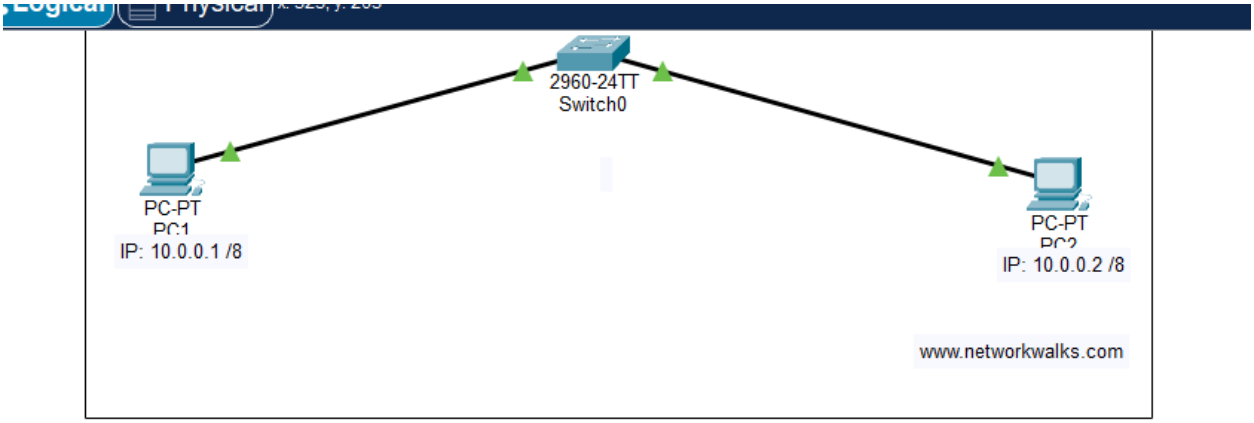
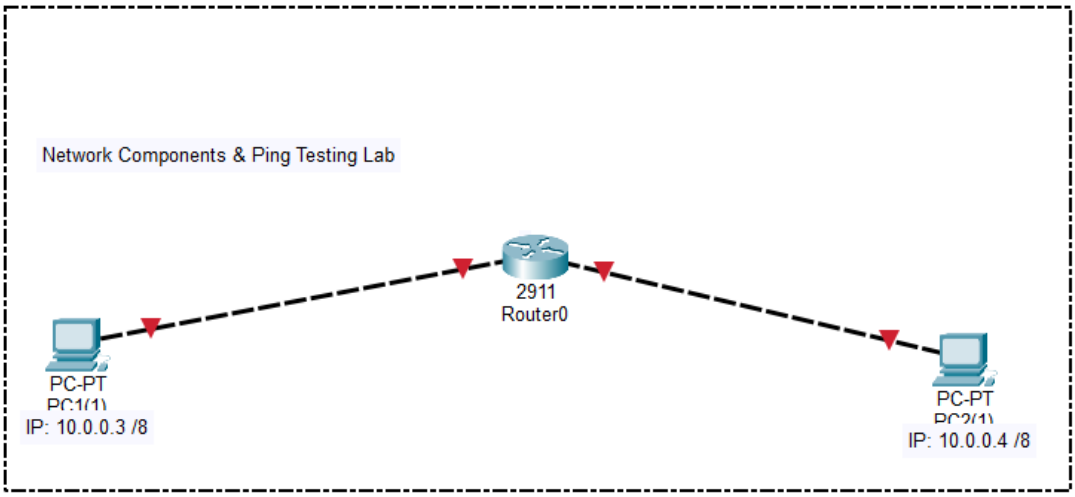
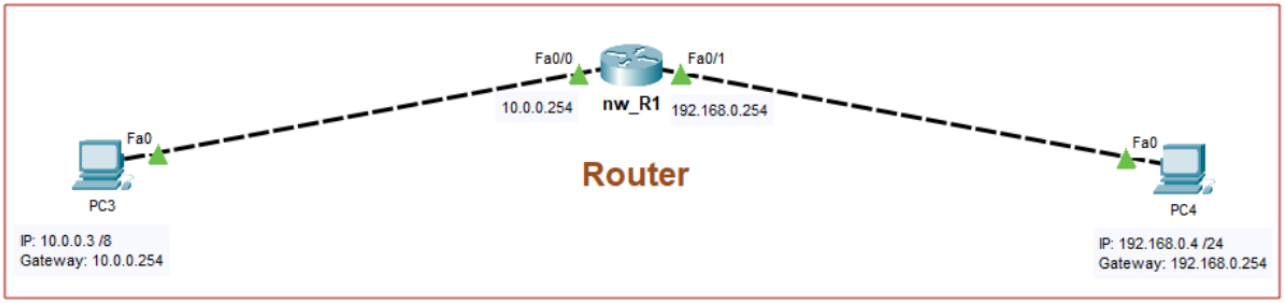
IP & MAC ADDRESS LAB

Topology Diagram

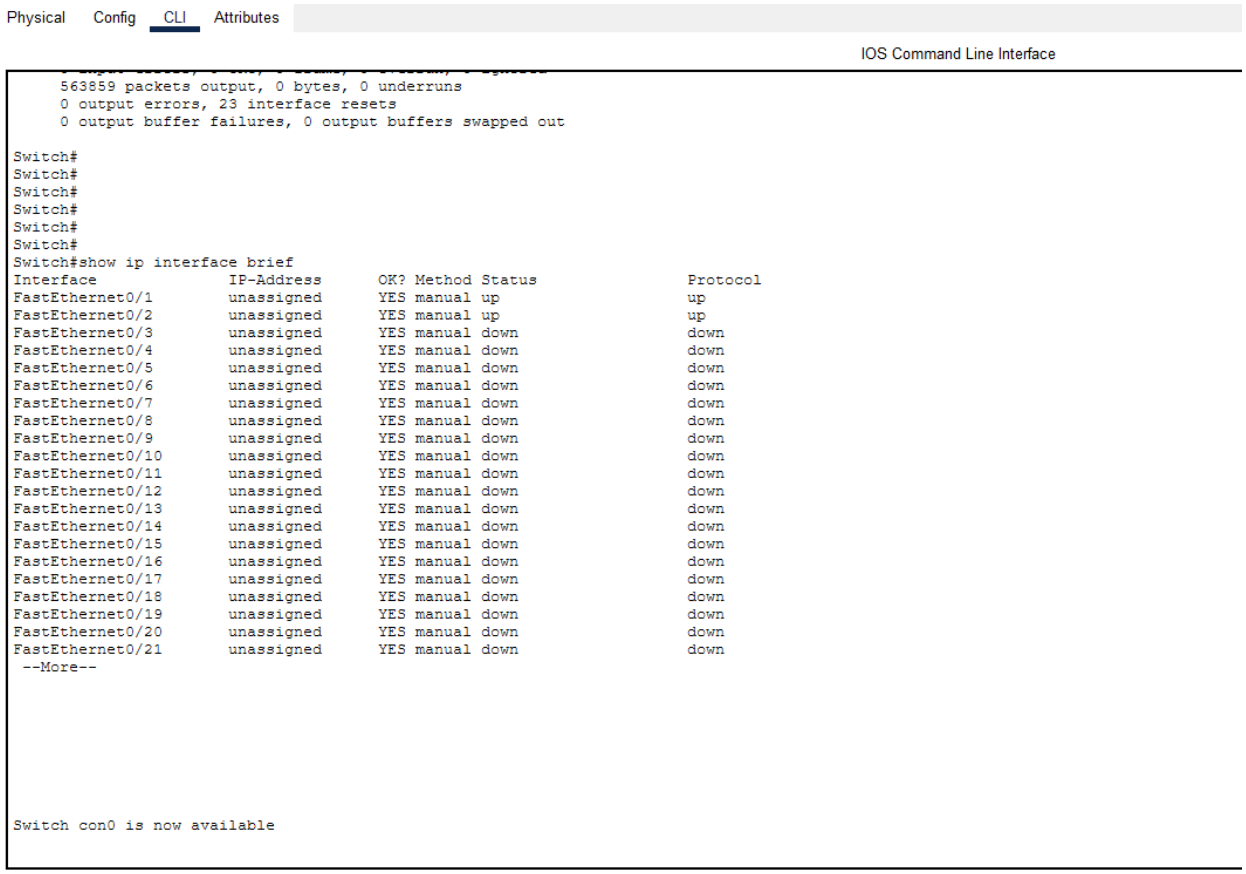






Task

1. Enable all relevant interfaces on Switch & Router (Fa0/0, Fa0/1, Fa0/2)



2. Assign IP addresses to PC's & Router interfaces (IP's are shown in topology)

```
%SYS-5-CONFIG_I: Configured from console by console

R1#
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.254      YES manual up                    up
GigabitEthernet0/1     192.168.0.254   YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#%IP-4-DUPADDR: Duplicate address 192.168.0.254 on GigabitEthernet0/1, sourced by 000A.F31C.DD61
%IP-4-DUPADDR: Duplicate address 192.168.0.254 on GigabitEthernet0/1, sourced by 000A.F31C.DD61
```

3. Test the connectivity between PC1-to-PC2 and PC3-to-PC4 using Ping command

```
inging 10.0.0.2 with 32 bytes of data:

eply from 10.0.0.2: bytes=32 time=33ms TTL=128
eply from 10.0.0.2: bytes=32 time=12ms TTL=128
eply from 10.0.0.2: bytes=32 time=9ms TTL=128
eply from 10.0.0.2: bytes=32 time=14ms TTL=128

ing statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
    Minimum = 9ms, Maximum = 33ms, Average = 17ms

:\>
```

4. Find what is the MAC address of switch port Fa0/1 using commands?

```
Switch>
Switch>
Switch>show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0040.0b9b.b276    DYNAMIC     Fa0/1
   1    0050.0f8c.6cb4    DYNAMIC     Fa0/2
Switch>
```

Copy

5. Find how many MAC entries are there in the Switch's MAC Table?

```
Switch>
Switch>
Switch>show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0040.0b9b.b276    DYNAMIC     Fa0/1
   1    0050.0f8c.6cb4    DYNAMIC     Fa0/2
Switch>
```

Copy

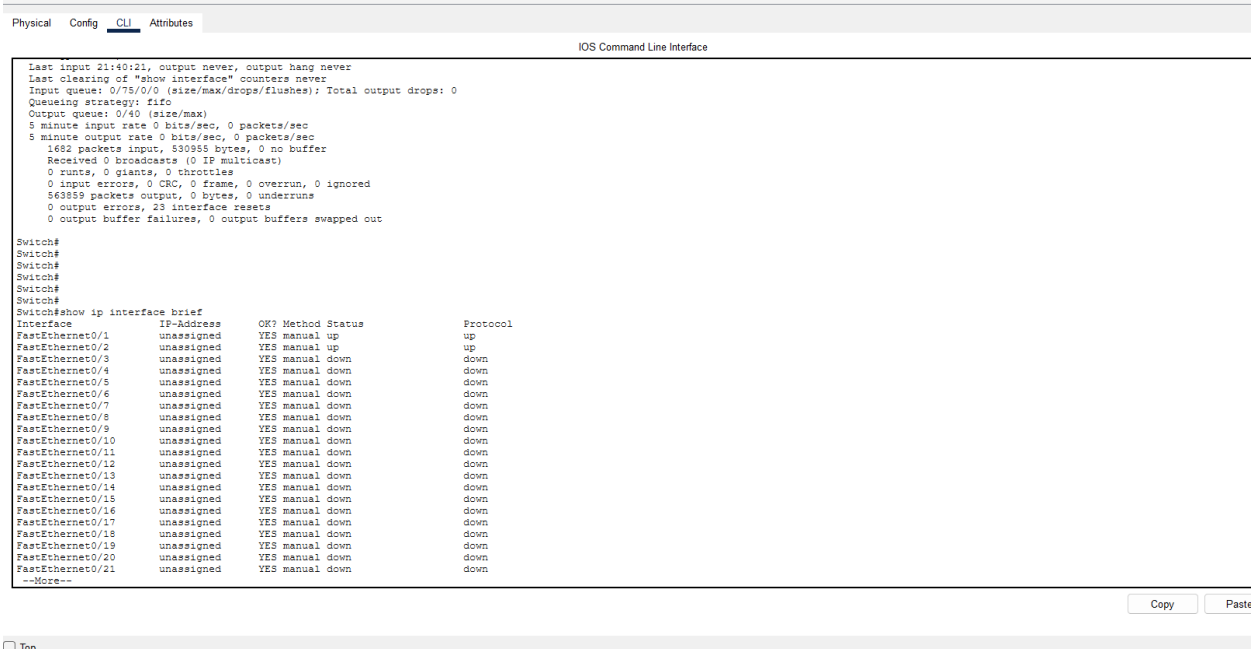
6. Find how many routes are there in the Routing table of nw_R1?

```
R1>
R1>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inte
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.0.254/32 is directly connected, GigabitEthernet0/0
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/1
L       192.168.0.254/32 is directly connected, GigabitEthernet0/1

R1>
```


Physical
Config
CLI
Attributes
IOS Command Line Interface
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
1682 packets input, 530955 bytes, 0 no buffer
Received 0 broadcasts (0 IP multicast)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
563859 packets output, 0 bytes, 0 underruns
0 output errors, 23 interface resets
0 output buffer failures, 0 output buffers swapped out
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES manual up up
FastEthernet0/2 unassigned YES manual up up
FastEthernet0/3 unassigned YES manual down down
FastEthernet0/4 unassigned YES manual down down
FastEthernet0/5 unassigned YES manual down down
FastEthernet0/6 unassigned YES manual down down
FastEthernet0/7 unassigned YES manual down down
FastEthernet0/8 unassigned YES manual down down
FastEthernet0/9 unassigned YES manual down down
FastEthernet0/10 unassigned YES manual down down
FastEthernet0/11 unassigned YES manual down down
FastEthernet0/12 unassigned YES manual down down
FastEthernet0/13 unassigned YES manual down down
FastEthernet0/14 unassigned YES manual down down
FastEthernet0/15 unassigned YES manual down down
FastEthernet0/16 unassigned YES manual down down
FastEthernet0/17 unassigned YES manual down down
FastEthernet0/18 unassigned YES manual down down
FastEthernet0/19 unassigned YES manual down down
FastEthernet0/20 unassigned YES manual down down
FastEthernet0/21 unassigned YES manual down down
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